

Handwritten Digit Recognition Using Neural Networks

MNIST Dataset • Convolutional Neural Network

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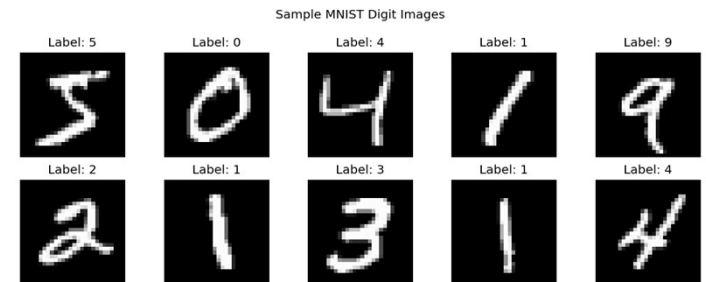
Agenda

- 1 Introduction & Problem Statement
- 2 Objectives
- 3 Dataset Overview
- 4 Methodology & Preprocessing
- 5 Model Architecture (CNN)
- 6 Training Process
- 7 Results & Evaluation
- 8 Sample Predictions
- 9 Conclusion & Future Scope

Introduction

Handwritten digit recognition is a foundational computer vision problem with wide real-world use — banking, postal automation, and form digitization.

This project builds a Convolutional Neural Network (CNN) to automatically classify handwritten digits (0–9) from the MNIST dataset.



Sample MNIST digits

Objectives

Understand CNN fundamentals for image classification

Preprocess and normalize the MNIST image dataset

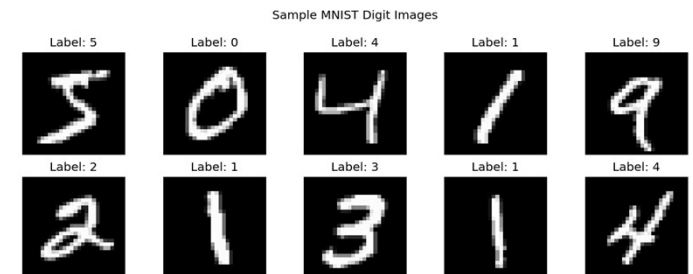
Design and train a CNN model

Evaluate performance using standard classification metrics

Build a working digit prediction system

Dataset Overview

Total Images	70,000
Training Samples	60,000
Testing Samples	10,000
Image Size	28 x 28 pixels (grayscale)
Classes	10 (digits 0–9)



Methodology & Preprocessing



Technologies: Python, TensorFlow, Keras, NumPy, Matplotlib, Scikit-learn

Model Architecture (CNN)

Conv2D (32 filters, 3x3, ReLU)

MaxPooling2D (2x2)

Conv2D (64 filters, 3x3, ReLU)

MaxPooling2D (2x2)

Flatten

Dense (128, ReLU) + Dropout (0.3)

Dense (10, Softmax)

225,034

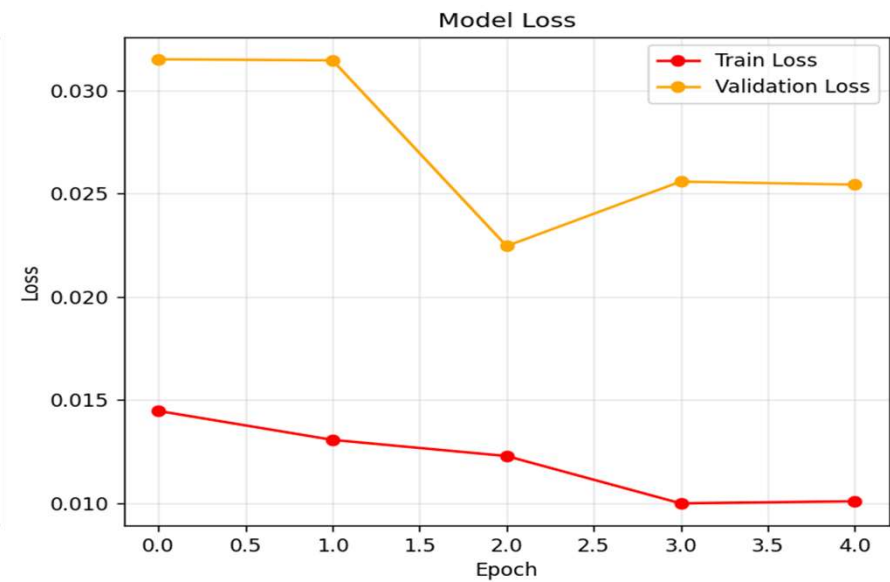
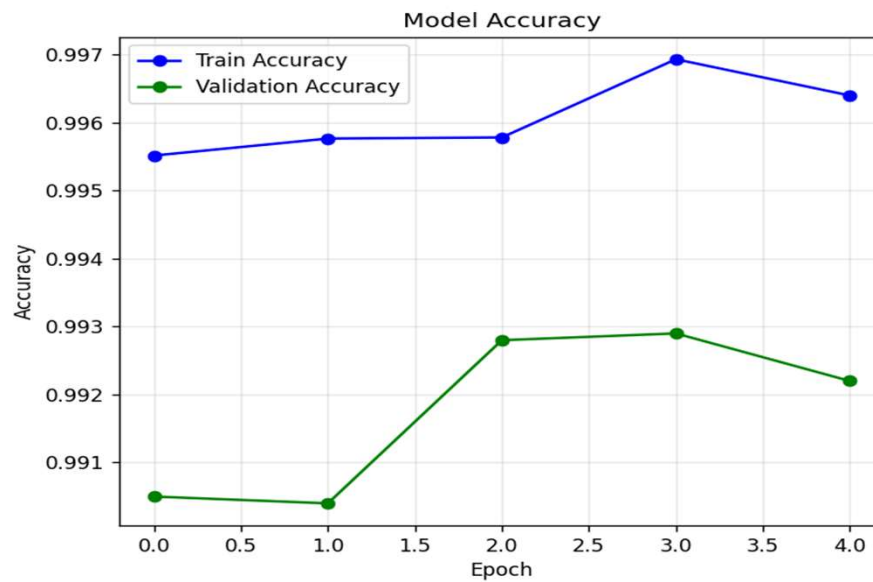
Total Trainable Parameters

Optimizer: Adam

Loss: Categorical Cross-Entropy

Epochs: 5 | Batch Size: 128

Training Process



Validation accuracy reached 99.24% by epoch 5, with steadily decreasing loss — no signs of overfitting.

Results & Evaluation

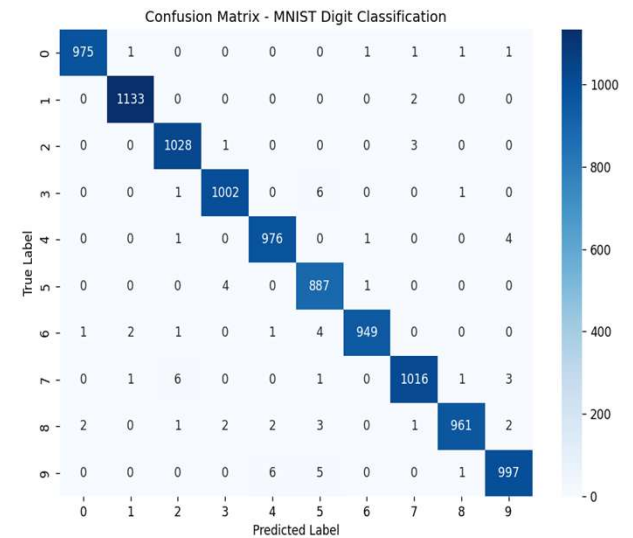
99.24%

Test Accuracy

0.0225

Test Loss

Precision, recall, and F1-score exceeded 0.98 across all 10 digit classes.



Sample Predictions

Model Predictions on Test Samples

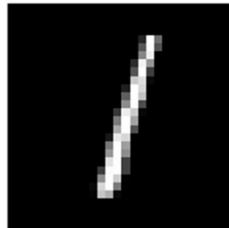
Pred: 7 | True: 7



Pred: 2 | True: 2



Pred: 1 | True: 1



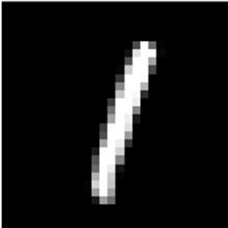
Pred: 0 | True: 0



Pred: 4 | True: 4



Pred: 1 | True: 1



Pred: 4 | True: 4



Pred: 9 | True: 9



Pred: 5 | True: 5



Pred: 9 | True: 9



Conclusion

- Built a CNN achieving 99.24% test accuracy on MNIST
- Covered the full ML pipeline: data → preprocessing → model → evaluation
- Demonstrated strong generalization with minimal overfitting
- Reinforces deep learning's relevance in real-world OCR systems

Future Scope

- Multi-digit and full alphanumeric recognition
- Real-time web/mobile deployment
- Deeper architectures for higher accuracy
- Integration into OCR pipelines (cheques, postal codes)

Thank You

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